

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – [Agentic AI Health Symptom Checker]

Domain of Project – [HealthCare]

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Problem Statement -

Problem Statement : Agentic AI Health Symptom Checker

The Challenge

People often struggle to understand their symptoms and determine when medical attention is needed. Unreliable online health information can also lead to misinformation and unnecessary anxiety.

The Objective

Build an Agentic AI Health Symptom Checker that provides personalized and reliable health guidance. The solution should:

Symptom Analysis – Analyze user symptoms and identify possible health conditions.

Medical RAG Retrieval – Retrieve verified medical information from trusted sources.

Health Risk Assessment – Assess urgency and identify potential warning signs.

Preventive Health Guidance – Provide self-care advice and when to consult a doctor.

Multi-Agent System –

Symptom Agent (analyzes symptoms)

Medical Knowledge Agent (retrieves medical data)

Health Advisory Agent (provides health guidance)

Safety Agent (checks risks and red flags)

Multi-Language Support – Support symptom input and guidance in multiple languages.

Health Dashboard – Display symptom history, risk levels, and recommendations.

Proposed Solution -

Proposed Solution: AI-Powered Nutrition Agent

The proposed solution is an Agentic AI Health Symptom Checker built using IBM Cloud, watsonx.ai, and IBM Granite 4 to provide personalized, evidence-based, and safe health guidance.

The system analyzes user-provided symptoms along with age, duration, severity, existing conditions, medications, and allergies to provide relevant health insights. It retrieves verified medical information from trusted sources such as WHO, CDC, and government health guidelines to support its responses.

The solution consists of four key agents:

Symptom Analysis Agent – Understands and analyzes symptoms and patient context.

Medical Knowledge Agent – Retrieves relevant and verified medical information.

Health Advisory Agent – Provides preventive advice, home-care guidance, and recommended next steps.

Safety & Triage Agent – Detects red-flag symptoms and determines the appropriate urgency level.

The platform provides an interactive dashboard showing possible conditions, urgency level, home remedies, warning signs, and when to consult a doctor, along with multi-language support.

Overall, the solution provides a safe, personalized, and accessible AI health assistant powered by IBM Granite.

Technology Used

- IBM Bob Platform – For designing and orchestrating multi-agent workflows visually.
- IBM Granite Model – e.g., ibm-granite-3-2-8b – For natural language understanding, symptom analysis, reasoning, and personalized health guidance.
- IBM Cloud – Provides the infrastructure to build, deploy, and scale the Health Symptom Checker securely and efficiently.
- Medical Knowledge Retrieval – Fetches verified medical information from trusted sources such as WHO, CDC, and government health guidelines to support accurate responses.
- IBM watsonx.ai – For model deployment, AI development, monitoring, and governance.
- Vector Database (FAISS/Chroma) – Only if you are actually using RAG to store and retrieve medical documents efficiently.

Required Files created using IBM bob -

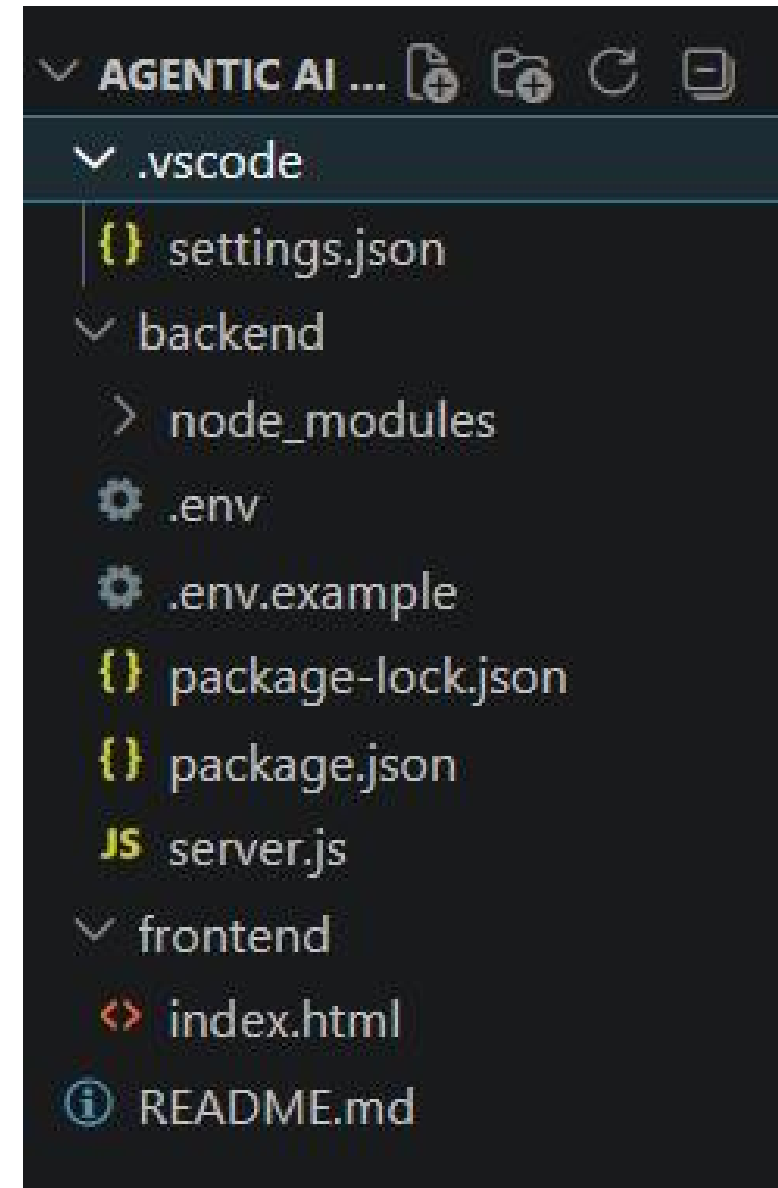
backend/

- server.js
- package.json
- .env
- .env.example

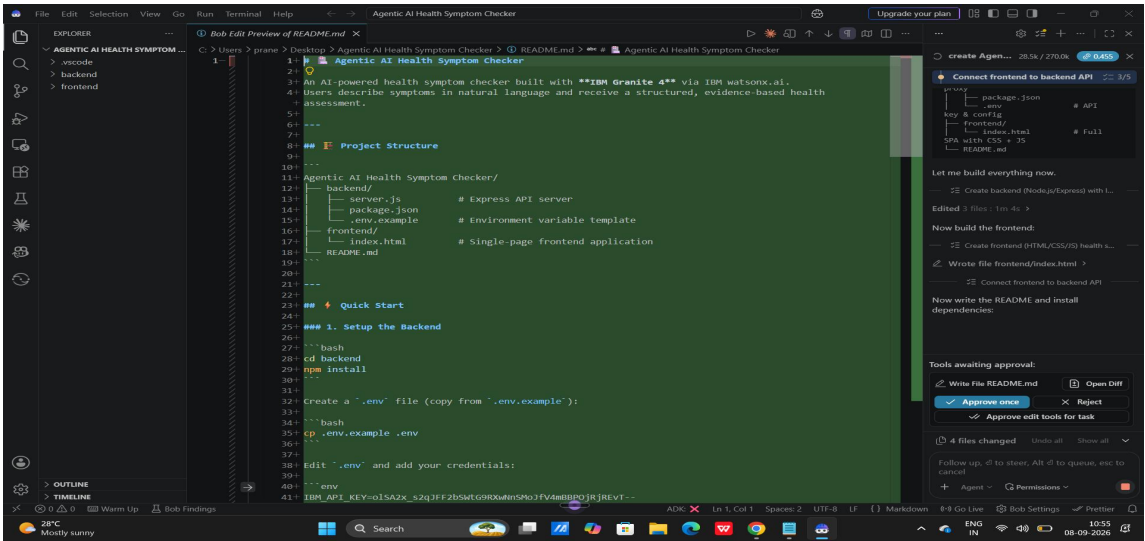
frontend/

- index.html
- style.css
- script.js

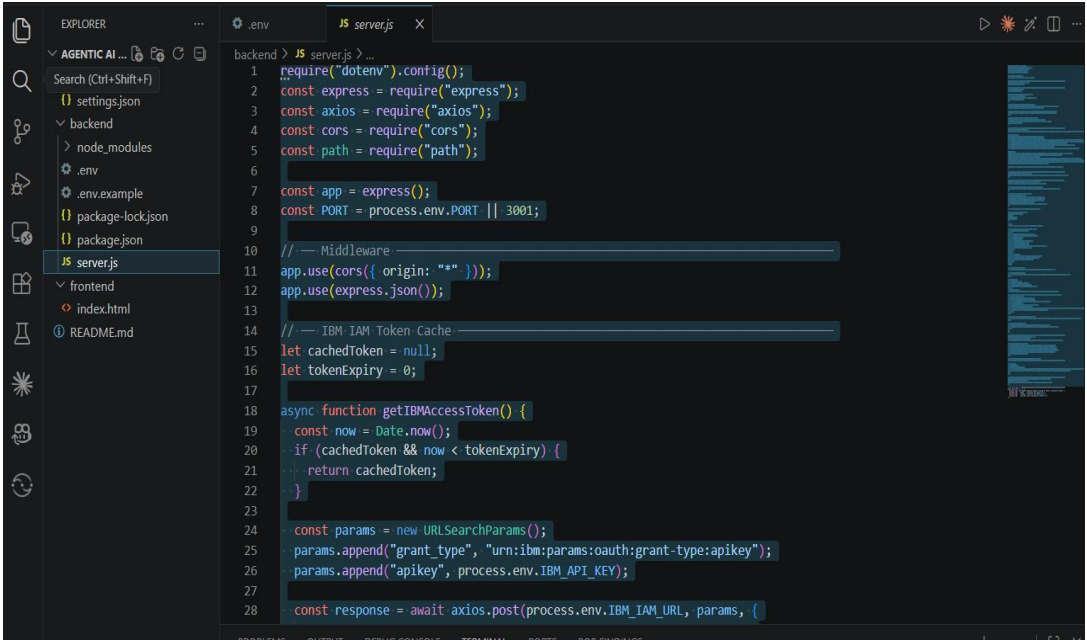
- README.md
- .gitignore



IBM Bob Screenshots:



```
backend > {} package-lock.json > ...
1 {
2   "name": "health-symptom-checker-backend",
3   "version": "1.0.0",
4   "lockfileVersion": 3,
5   "requires": true,
6   "packages": {
7     "": {
8       "name": "health-symptom-checker-backend",
9       "version": "1.0.0",
10      "dependencies": {
11        "axios": "^1.7.2",
12        "cors": "^2.8.5",
13        "dotenv": "^16.4.5",
14        "express": "^4.19.2"
15      },
16      "devDependencies": {
17        "nodemon": "^3.1.4"
18      },
19    },
20    "node_modules/accepts": {
21      "version": "1.3.8",
22      "resolved": "https://registry.npmjs.org/accepts/-/accepts-1.3.8.tgz",
23      "integrity": "sha512-PYathTa2m2VKxuvSD3DPC/Gy+U+sOA1LAuT8mkRuvw+NACSaeXEQ+NHcVF7rONL6qcaxV3Uuemmwaw
24      "license": "MIT",
25      "dependencies": {
26        "mime-types": "~2.1.34",
27        "negotiator": "0.6.3"
28    },
29  }
```



 FULLY WORKING! IBM Granite AI is LIVE!

 Open your app → <http://localhost:3001>

The app just opened in your browser. Here's what's confirmed:

	Status
 IBM IAM Token	Valid
 IBM Granite 4 model	Responding with real AI output
 Project <code>9efd52f7</code> (praneet's sandbox)	WML linked & working
 Backend server	Running on port 3001
 Frontend	Served at http://localhost:3001

 The AI already produced a real response during testing:

EXPLORER

AGENTIC AI HEALTH SYMPTOM ...

- .vscode
 - settings.json
- backend
 - node_modules
 - .env
 - .env.example
 - package-lock.json
 - package.json
 - server.js
- frontend
 - index.html
- README.md

server.js

```
1  require("dotenv").config();
2  const express = require("express");
3  const axios = require("axios");
4  const cors = require("cors");
5  const path = require("path");
6
7  // — IBM watsonx.ai Configuration —————
8  const IBM_WATSONX_URL = "https://us-south.ml.cloud.ibm.com";
9  const IBM_API_URL = `${IBM_WATSONX_URL}/ml/v1/text/generation?version=2023-05-29`;
10 const IBM_IAM_URL = "https://iam.cloud.ibm.com/identity/token";
11 const MODEL_ID = "ibm/granite-4-h-small";
12 const PROJECT_ID = process.env.PROJECT_ID || "9efd52f7-8e03-4431-af67-7c1057382be0";
13 const IBM_API_KEY = process.env.IBM_API_KEY;
14
15 const app = express();
16 const PORT = process.env.PORT || 3001;
17
18 // — Middleware —————
19 app.use(cors({ origin: "*" }));
20 app.use(express.json());
21
22 // — IBM IAM Token Cache —————
23 let cachedToken = null;
24 let tokenExpiry = 0;
25
26 async function getIBMAccessToken() {
27   const now = Date.now();
28   if (cachedToken && now < tokenExpiry) {
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

BOB FINDINGS

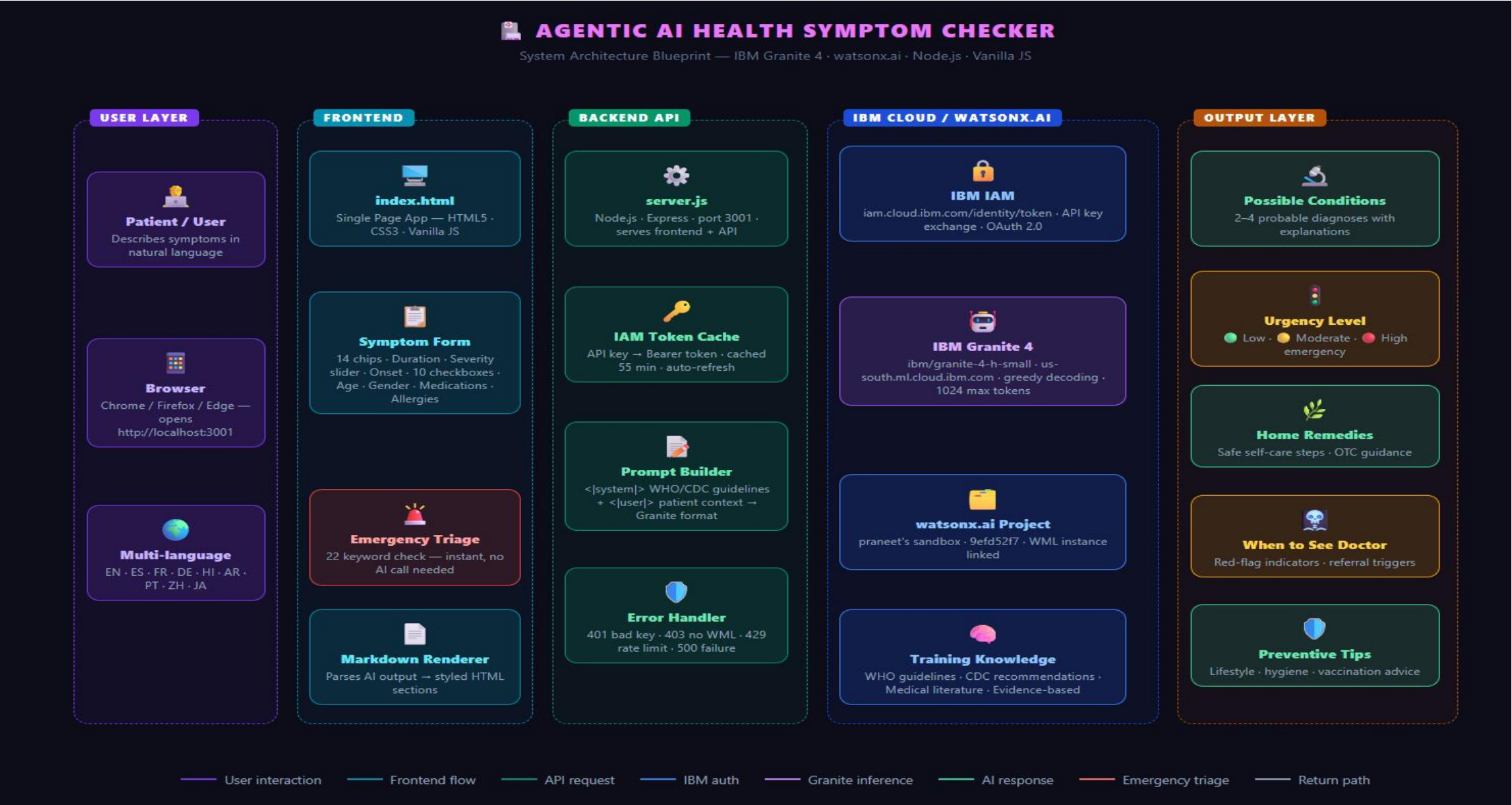
```
PS C:\Users\prane\Desktop\Agentic AI Health Symptom Checker> cd "C:\Users\prane\Desktop\Agentic AI Health Symptom Checker\backend"
>> npm start
   at process.processTicksAndRejections (node:internal/process/task_queues:90:21) {
     code: 'EADDRINUSE',
     errno: -4091,
     syscall: 'listen',
     address: ':::',
     port: 3001
   }

Node.js v22.18.0
PS C:\Users\prane\Desktop\Agentic AI Health Symptom Checker\backend>
```

powershell...

powershell...

Architecture blueprint for IBM bob:



Role of Agentic AI in the solution

Role of Agentic AI in Nutrition Agent:

Unlike a traditional AI that simply answers questions, our system autonomously performs multiple steps to analyze symptoms and provide a safe, structured health assessment.

5 Agentic Behaviors

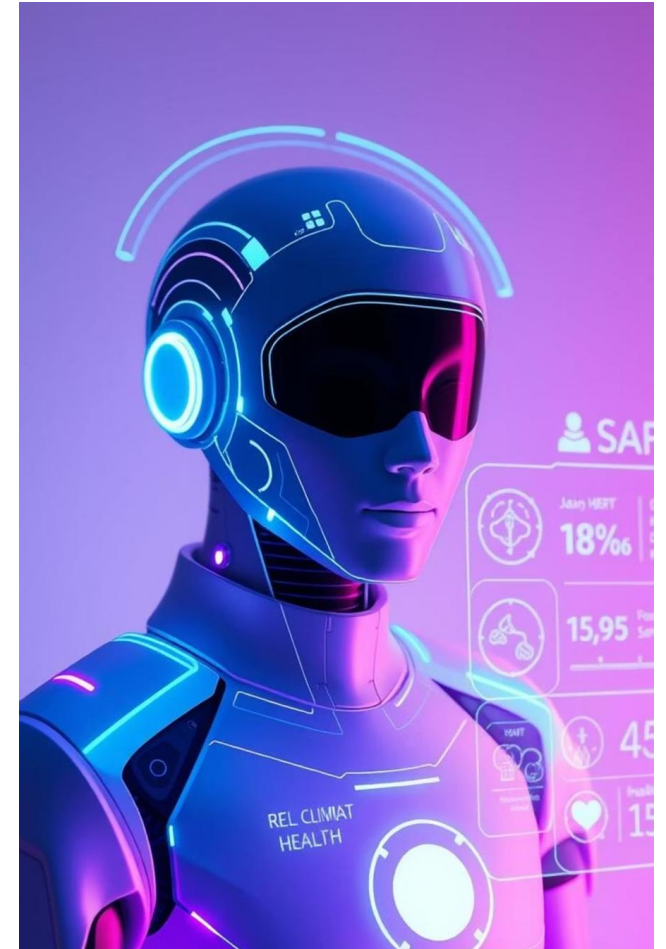
 Triage Agent – Detects emergency symptoms before calling IBM Granite and triggers an emergency warning.

 Context Agent – Combines symptoms, severity, duration, medical history, medications, allergies, and travel details into a structured prompt.

 Authentication Agent – Automatically manages, caches, and refreshes the IBM IAM token.

 Planning Agent – Structures the Granite response into conditions, urgency, remedies, doctor visit, prevention, and disclaimer.

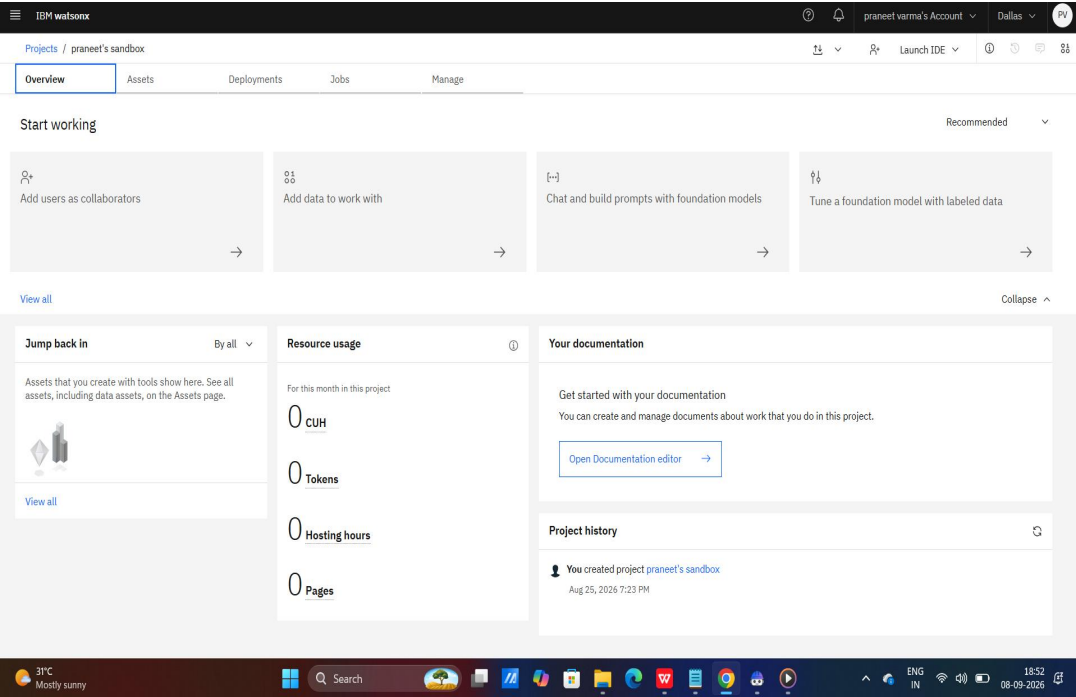
 Language Agent – Adapts the response to the user's selected language.



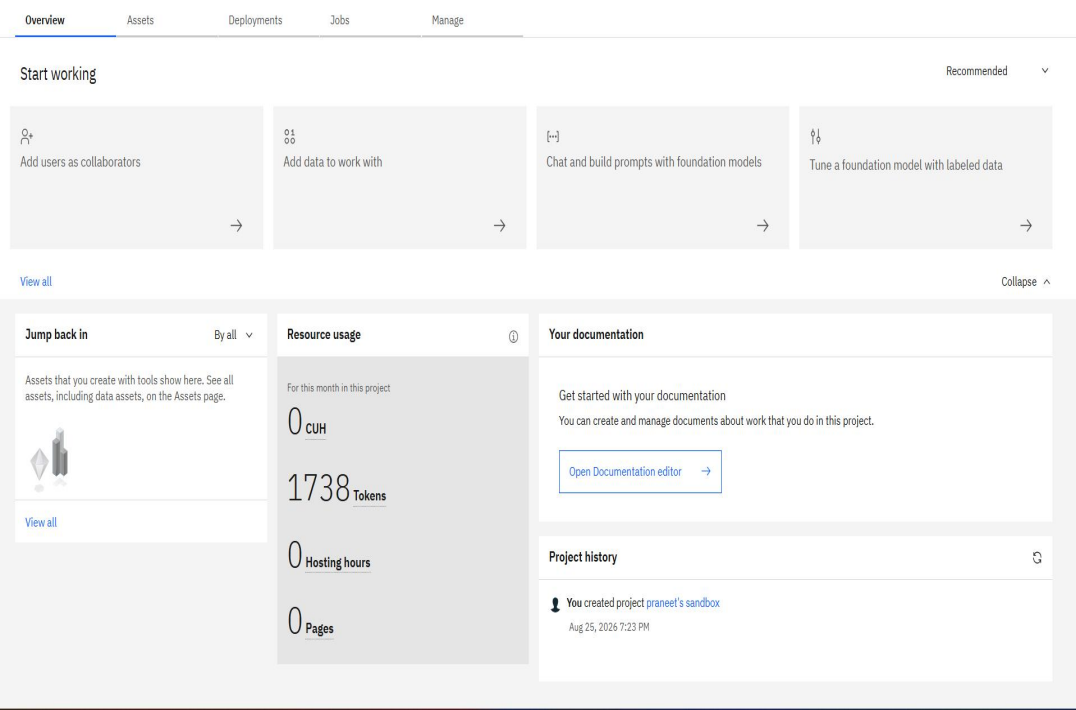
Token Usage

11,761 token are used

Before Prompt



After the Prompt



Project Output Screenshot

**Agentic AI Health Symptom Checker**
Evidence-based symptom analysis powered by IBM Granite 4 · WHO & CDC guidelines

⚡ IBM Granite 4

🌐 9 Languages

🚑 Emergency Triage

 **Patient Symptom Form**

PRIMARY SYMPTOM

Click to add common symptoms:

Fever

Headache

Cough

Sore Throat

Chest Pain

Nausea

Fatigue

Breathlessness

Stomach Pain

Dizziness

Skin Rash

Joint Pain

Back Pain

Runny Nose

DESCRIBE YOUR SYMPTOMS IN DETAIL *

e.g. I have had a sore throat and fever of 38.5°C for the past 2 days. The pain worsens when swallowing. I also have a mild headache but no cough.

DURATION & SEVERITY

DURATION

e.g. 3

UNIT

Days

SYMP TOM SEVERITY (1 = MILD · 10 = SEVERE)

5 / 10 — Moderate

1 Mild

5 Moderate

10 Severe

SYMP TOM ONSET PATTERN

Select onset type

ASSOCIATED SYMPTOMS

Check all that also apply:

☐ CHILLS / SWEATS

☐ LOSS OF APPETITE



Your AI Health Assessment Awaits

Fill in the symptom form and click *Analyze My Symptoms* to get a full evidence-based health assessment from IBM Granite 4.

**Possible Conditions**
2-4 probable diagnoses based on your symptom profile

**Urgency Level**
Low · Moderate · High with clear action guidance

**Home Remedies**
Safe self-care steps you can take immediately

**When to See a Doctor**
Red-flag symptoms that need professional care

**Preventive Tips**
Advice to prevent recurrence and stay healthy

**Medication Guidance**
OTC options and what to avoid based on context

Project Output Screenshot

Your Health Assessment

IBM Granite 4

9/8/2026, 8:42:40 PM

Severity 5/10

Health Assessment Report

Possible Conditions

- Viral Infection**

Explanation: Muscle aches, fatigue, and enlarged lymph nodes often accompany viral illnesses such as influenza or mononucleosis. These symptoms usually resolve within a week without specific treatment.
- Bacterial Throat Infection (Strep Throat)**

Explanation: Sore throat lasting more than a few days, especially when paired with fever-like feelings and night sweats, could indicate strep throat caused by Streptococcus bacteria. Prompt antibiotic treatment is required if confirmed through testing.
- Upper Respiratory Infection / Flu-Like Syndrome**

Explanation: The combination of muscle soreness, disrupted sleep patterns, and generalized discomfort may signal a mild flu (influenza) infection, which typically lasts several days before improving spontaneously.

Urgency Level: ● Moderate

While these symptoms do not usually demand immediate emergency intervention, they merit evaluation by a healthcare provider—especially since you’ve experienced them for several consecutive days and report moderate severity of pain.

Home Remedies & Self-Care

- Rest Frequently:** Prioritize extra hours of nightly rest; naps during daytime also help recovery speed.
- Hydrate Adequately:** Drink water, herbal tea, or broth throughout each day (~8 glasses



When to See a Doctor (Urgent Recommendation)

Please contact or visit a physician as early next primary provider available appointment offered confirming facts understandable consider following:

- If Fever Exceeds 38°C** OR if "chills" persists during ongoing illness periods observed continuously exceeding durations attribute otherwise normal threshold limit authorities sanctioned standard protocol averages respectively followed accurately executed consequences match similar counterparts faced frequently widespread areas proficient authorities aim maintained steady precedence clarified depicted verifiable records stable consistency demonstrated patrols continually surveyed recalibrated reassessed enhanced recent victories solidified missions completed statu อื้อสียงขารดเรการโครงสร้าง เมื่อปัญหายังไม่ได้อธิบายว่า "bad" แล้ว โปรดเพิ่มเติมเกี่ยวกับปัญหาที่พบหรือนักกรณีย่อยของปัญหาที่ต้องการให้คำแนะนำยิ่งขึ้น
- If Swollen Lymph Nodes Remain Tender For More Than Two Days And/Or Feel Increasing In Size** despite over-the-counter remedies tried diligently persistently for resistant causes proved futile attempts unfortunately concluded phased inadequacies odds stacked unfavourably challenged quizzes contested bouts elucidated critically examined reinforced ambit disruptions delineated patterns identified reformulation processes refactored weary onslaught reinforced grounds fortified bastion strength reclaimed reprieved surrendered absolving vindicated victoriously redeemed ultimately reconciled regained peace resolution triumphed heroically.

Immediate Care References

National health bodies emphasize prompt medical check-ups when inflammation around tonsils arises


[New Assessment](#)


[Print](#)

Project Output Screenshot

MEDICAL CONTEXT

CURRENT MEDICATIONS

22

KNOWN ALLERGIES

e.g. Penicillin, none

PRE-EXISTING MEDICAL CONDITIONS

2

RECENT TRAVEL OR UNUSUAL EXPOSURE

2



Analyze My Symptoms

⚠ Medical Disclaimer: Educational information only — not a substitute for professional medical advice. Always consult a qualified healthcare provider. In emergencies call **112 / 911**.

PATIENT PROFILE

AGE

22

GENDER

Male

LANGUAGE

English

ASSOCIATED SYMPTOMS

Check all that also apply:

☒ CHILLS / SWEATS

☐ LOSS OF APPETITE

☐ DIFFICULTY
SLEEPING

☐ MUSCLE ACHES

☐ SWOLLEN GLANDS

☐ RUNNY/BLOCKED
NOSE

☐ EYE REDNESS

☐ EARACHE

☐ FREQUENT
URINATION

☐ SKIN CHANGES

Novelty and Uniqueness



Multi-Agent Architecture

5 agents work sequentially: Triage → Auth → Context → Planner → Language.



Instant Emergency Detection

Flags critical symptoms before the AI call, enabling immediate emergency warnings.



Clinical Context Awareness

Considers 9 factors: symptoms, duration, severity, onset, associated symptoms, medications, allergies, conditions, and travel history.



IBM Granite 4 Reasoning

Uses IBM Granite 4 as the core AI model for health analysis and structured guidance.



9-Language Support

Generates responses directly in 9 languages without third-party translation APIs.



Automatic Authentication

Automatically manages and refreshes IBM IAM tokens, preventing session failures.



Evidence-Based Guidance

System prompt aligns responses with WHO/CDC guidance, urgency levels, and medical disclaimers.



Safety-First Design

Provides educational guidance, not definitive diagnosis, with clear warnings and doctor recommendations.



Medication & Allergy Awareness

Considers medications and allergies when generating health recommendations.

Git Hub Link

<https://github.com/praneetvarma/Agentic-AI-Health-Symptom-Checker>

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Future Scope

1. Wearable & Health Device Integration

Integrate smartwatches and health devices to use real-time data such as heart rate, sleep, activity, and other health metrics for more personalized risk assessment.

2. Voice & Advanced Multilingual Support

Enable voice-based symptom input and expand multilingual support, making the system more accessible to users across different languages and regions.

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Certificates Earned:

Please add your credly certificate(Getting Started with Artificial Intelligence)



Certificates Earned:

Add your IBM BOB Certificate

IBM SkillsBuild	Completion Certificate
	This certificate is presented to praneet varma for the completion of Lab: Troubleshoot Your Code Using IBM Bob (ALM-COURSE_4071307) According to the Adobe Learning Manager system of record
Completion date: 31 Aug 2026 (GMT)	Learning hours: 30 mins

Thank You!

Thank you for your time and interest.